Table 2. Before seeing any label, both adjudicators' own ratings agreed with ARBITER far more than with the unassisted model or the published review

Comparison	Weighted κ	Raw agreement
Between adjudicators		
Independent ratings	0.59 (0.48–0.68)	114/154 (74.0%; 68.2–79.9)
Ratings chosen after seeing labels	0.69 (0.57–0.79)	129/154 (83.8%; 78.6–88.9)
Adjudicator 1 own rating compared with …		
Published	−0.08 (−0.17 to 0.01)	34/154 (22.1%; 15.8–28.5)
ARBITER	0.52 (0.41–0.62)	107/154 (69.5%; 62.7–76.0)
Unassisted	0.22 (0.13–0.30)	69/154 (44.8%; 38.1–51.6)
Adjudicator 2 own rating compared with …		
Published	−0.08 (−0.18 to 0.03)	42/154 (27.3%; 20.4–34.3)
ARBITER	0.67 (0.56–0.76)	120/154 (77.9%; 71.7–84.0)
Unassisted	0.31 (0.19–0.42)	76/154 (49.4%; 42.2–56.5)
Post hoc: paired agreement difference (percentage points)		
Adjudicator 1	—	24.7 (16.2–32.9)
Adjudicator 2	—	28.6 (19.8–37.4)

All comparisons use the 154 domain ratings reviewed because at least one arm differed from the published label. These comparisons cannot estimate agreement or published-label accuracy in the full benchmark. Adjudicators used no AI; each independent rating was made and frozen before the adjudicator saw any label. They then saw only three labels in an order randomised per cell, without rationales or quotations. Linearly weighted Cohen's κ; AC2 is in eTable 13; raw agreement is n/N (x.x%; 95% CI a–b). Post hoc paired differences are ARBITER minus Unassisted agreement with the same independent rating, in percentage points. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018.